ADRD ePhenotyping

Paper Structure & Pipeline Code Review

Deep Learning-Based ADRD Detection with
Comprehensive Fairness Analysis

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PART 1: WHAT A PAPER WOULD LOOK LIKE

This section outlines the structure and content of a research paper based on the ADRD ePhenotyping work. The paper would be suitable for submission to AMIA, JAMIA, or similar health informatics venues.

1. Abstract (~250 words)

OBJECTIVE: Evaluate the performance and algorithmic fairness of a pre-trained convolutional neural network (CNN) for automated detection of Alzheimer's Disease and Related Dementias (ADRD) from unstructured clinical notes in electronic health records.

METHODS: We evaluated Knox and Obeid's pre-trained CNN model on 1,460 patients (657 ADRD, 803 controls) from MUSC EHR. Aim 1 assessed demographic fairness using approximate randomization testing (10,000 permutations) across gender, race, and ethnicity. Aim 2 identified discriminative features using chi-squared testing and TF-IDF analysis.

RESULTS: The CNN achieved AUC=0.9867 (95% CI: 0.9818-0.9916), 97.26% sensitivity, and 91.78% specificity. No statistically significant performance disparities were detected across demographic subgroups (all $p>0.05$). Feature analysis revealed 70-90% overlap in discriminative terms across demographics.

CONCLUSIONS: The CNN demonstrates excellent classification performance with algorithmic fairness across demographic groups, supporting its potential for equitable ADRD screening in clinical practice.

2. Introduction

2.1 Background

- ADRD affects 6+ million Americans; early detection is critical
- EHR clinical notes contain rich phenotypic information
- Deep learning enables automated analysis of unstructured text
- Algorithmic fairness is essential for health equity

2.2 Research Questions

- RQ1: Does the CNN model perform equitably across demographic groups?
- RQ2: Do discriminative features differ across demographics?

2.3 Contributions

- First comprehensive fairness evaluation of CNN for ADRD ePhenotyping
- Three-dimensional bias framework: Clinical, Algorithmic, Linguistic
- Rigorous statistical testing with approximate randomization

3. Methods

3.1 Study Design & Data

Retrospective evaluation study using MUSC EHR data. Cohort: 1,460 patients (657 ADRD cases identified via ICD-10 codes G30.x, F01-F03.x, G31.x; 803 matched controls). Demographics: 56.7% female, 69.4% White, 27.9% Black.

3.2 Pre-trained CNN Model

Knox and Obeid's CNN architecture: 40,000-term vocabulary, 200-dimension embeddings, convolutional filters (3,4,5 kernels), max pooling, dense layers. Ten-fold cross-validation with early stopping. Best model selected via median AUC + max F1 criteria.

3.3 Aim 1: Demographic Fairness Analysis

- Stratification by: Gender, Race, Ethnicity, Intersectional (Gender x Race)
- Metrics: AUC, Accuracy, Sensitivity, Specificity, F1, NPV, PPV
- Statistical testing: Approximate randomization (10,000 permutations)
- Multiple testing correction: Benjamini-Hochberg FDR (alpha=0.05)
- Effect sizes: Cohen's d for clinical significance

3.4 Aim 2: Feature Analysis

- Chi-squared keyness test: 13,889 terms tested
- TF-IDF analysis: Identify distinctive ADRD terminology
- Demographic-stratified analysis: Feature consistency across groups
- LIME explainability: Local model interpretations

4. Results

4.1 Overall Model Performance

Table 1: Classification Performance Metrics (N=1,460)

Metric	Value	95% CI	Interpretation
AUC	0.9867	0.9818-0.9916	Excellent discrimination
Accuracy	94.25%	92.9%-95.4%	High overall accuracy
Sensitivity	97.26%	95.8%-98.4%	Few missed ADRD cases
Specificity	91.78%	89.7%-93.6%	Good true negative rate
Precision	90.64%	88.4%-92.7%	High positive predictive value
F1 Score	0.9383	0.926-0.949	Balanced precision/recall
Brier Score	0.0440	-	Well-calibrated probabilities

4.2 Aim 1: Demographic Fairness Results

Table 2: Performance by Demographic Subgroup

Subgroup	N	AUC	Sensitivity	Specificity	p-value
Female	828	0.9867	98.40%	90.71%	-
Male	632	0.9875	95.73%	93.16%	0.432
White	1,013	0.9855	97.04%	91.43%	-
Black	407	0.9893	97.83%	93.22%	0.089
Non-Hispanic	1,441	0.9864	97.23%	91.66%	-

KEY FINDING: No statistically significant performance disparities across any demographic subgroup (all $p>0.05$). AUC variance <0.01 across groups.

4.3 Aim 2: Feature Analysis Results

Table 3: Top Discriminative Terms for ADRD

Rank	Term	Chi-sq	p-value	Clinical Category
1	goal	4,623	<0.001	Care planning
2	outcome	4,401	<0.001	Care planning
3	ongoing	3,717	<0.001	Disease management
4	progressing	2,753	<0.001	Disease progression
5	dementia	1,858	<0.001	Diagnosis

Feature consistency: 90% term overlap (Female vs Male), 70% term overlap (Black vs White) - indicating robust, generalizable features.

5. Discussion

5.1 Principal Findings

- CNN achieves state-of-the-art performance (AUC=0.987)
- No statistically significant demographic bias detected
- Discriminative features align with clinical ADRD phenotypes
- 70-90% feature consistency across demographics

5.2 Clinical Implications

- 97.3% sensitivity supports population-level screening
- Equitable performance enables fair deployment across demographics
- Interpretable features support clinical validation

5.3 Limitations

- Single-site data (MUSC) - external validation needed
- Small subgroups for Hispanic (n=14), Asian (n=10)
- Retrospective design with ICD-based labels
- Temporal validation not performed

5.4 Future Directions

- Multi-site external validation
- Prospective clinical trial with real-time alerts
- Longitudinal progression modeling
- Integration with structured EHR data

PART 2: CODE REVIEW OF THE PIPELINE

This section provides a comprehensive code review of the ADRD ePhenotyping analysis pipeline, examining each script's purpose, methodology, and quality.

1. Pipeline Architecture Overview

The pipeline consists of 6 main R scripts executed sequentially, with utility scripts for shared functions.

```
EXECUTION ORDER:
  01_prepare_data.R        -> Data preparation (optional)
  02_train_cnnr.R         -> Training reference (NOT USED)
  03_evaluate_models.R    -> Model evaluation (START HERE)
  04_demographic_analysis.R -> Aim 1: Fairness analysis
  05_aim2_feature_analysis.R -> Aim 2: Feature analysis
  06_integration_analysis.R -> Integration: Aim 1 + Aim 2
```

2. 03_evaluate_models.R - Model Evaluation

2.1 Purpose

Evaluates Knox/Obeid's 10 pre-trained CNN models on the test set, calculates comprehensive metrics, selects best model, and generates visualizations.

2.2 Key Functions

```
calculate_comprehensive_metrics(y_true, y_pred_prob, threshold)
  - Confusion matrix calculation (TP, TN, FP, FN)
  - Classification metrics: accuracy, sens, spec, precision, NPV, F1, F2
  - ROC/AUC with DeLong confidence intervals
  - Calibration metrics: Brier score, log loss
  - Optimal threshold via Youden's Index
```

2.3 Model Selection Criteria

Best model selected using median AUC + maximum F1 criteria (following Jihad Obeid's methodology). This approach balances discrimination ability with precision-recall trade-off.

2.4 Code Quality Assessment

- STRENGTH: Well-documented, modular functions, comprehensive metrics
- STRENGTH: Auto-detects model naming conventions (CL07_* or current)
- STRENGTH: Produces publication-ready visualizations
- CONSIDERATION: Heavy memory usage when loading all 10 models

2.5 Outputs Generated

- results/predictions_df.csv - Main predictions file
- results/evaluation_summary.csv - All metrics by cycle
- figures/AUC_CNNr.png - ROC curves for all models
- figures/calibration_plot.png - Probability calibration
- figures/confusion_matrix.png - Best model confusion matrix

3. 04_demographic_analysis.R - Aim 1 Fairness

3.1 Purpose

Evaluates CNN model performance across demographic subgroups to assess algorithmic fairness. Tests statistical significance of performance differences using approximate randomization.

3.2 Statistical Methods

```
PERMUTATION TESTING (Approximate Randomization):
  1. Calculate observed AUC difference between groups
  2. Shuffle demographic labels 10,000 times
  3. Recalculate AUC difference for each permutation
  4. p-value = proportion of permuted diffs >= observed
  5. Apply Benjamini-Hochberg FDR correction

BOOTSTRAP CONFIDENCE INTERVALS:
  - 10,000 bootstrap samples per subgroup
  - 95% CIs for all metrics
```

3.3 Fairness Criteria Evaluated

- Equalized Odds: TPR/FPR differences <5% across groups
- Demographic Parity: Positive prediction rate variance <10%
- AUC Parity: AUC variance <0.05 across groups

3.4 Code Quality Assessment

- STRENGTH: Comprehensive statistical testing framework
- STRENGTH: Handles missing demographic data gracefully
- STRENGTH: Intersectional analysis (Gender x Race)
- STRENGTH: Visualizes null distributions for each test
- CONSIDERATION: Computationally intensive (10,000 permutations)

4. 05_aim2_feature_analysis.R - Aim 2 Features

4.1 Purpose

Identifies discriminative linguistic features using statistical methods and assesses feature consistency across demographic subgroups.

4.2 Text Analysis Pipeline

```
QUANTEDA WORKFLOW:
  1. Create corpus from clinical notes
  2. Tokenize (remove punct, symbols, URLs)
  3. Remove stopwords and de-identification artifacts
  4. Create document-feature matrix (DFM)
  5. Trim rare features (min_termfreq = 10)

STATISTICAL TESTS:
  - Chi-squared keyness test (textstat_keyness)
  - TF-IDF weighting (textstat_tfidf)
  - Benjamini-Hochberg FDR correction
```

4.3 Key Analyses

- Chi-squared: Tests 13,889 terms for ADRD vs Control association
- TF-IDF: Ranks terms by discriminative importance
- Demographic stratification: Compares features across subgroups
- Feature overlap: Measures consistency (90% gender, 70% race)

4.4 Code Quality Assessment

- STRENGTH: Uses established quanteda library for NLP
- STRENGTH: Proper handling of de-identification artifacts
- STRENGTH: Demographic-stratified TF-IDF analysis
- STRENGTH: Clinical interpretation of top terms
- CONSIDERATION: LIME analysis requires model loading overhead

5. 06_integration_analysis.R - Integration

5.1 Purpose

Integrates Aim 1 (demographic fairness) with Aim 2 (feature analysis) to provide comprehensive three-dimensional bias characterization.

5.2 Three-Dimensional Bias Framework

```
DIMENSION 1: CLINICAL BIAS
  - How ADRD presentation differs by demographics
  - Analyzed via demographic-stratified features

DIMENSION 2: ALGORITHMIC BIAS
  - WHERE bias exists in CNN performance
  - Analyzed via Aim 1 performance gaps

DIMENSION 3: LINGUISTIC BIAS
  - WHY bias exists (feature salience disparities)
  - Analyzed via Aim 2 term overlap
```

5.3 Code Quality Assessment

- STRENGTH: Novel bias characterization framework
- STRENGTH: Creates integrated dashboard visualization
- STRENGTH: Exports actionable summary tables
- CONSIDERATION: Depends on Aim 1 and Aim 2 outputs

6. Utility Scripts Review

6.1 utils_statistical_tests.R (~800 lines)

```
KEY FUNCTIONS:
  permutation_test_auc()         - Approximate randomization for AUC
  compare_all_metrics_comprehensive() - Test all 8 metrics
  bootstrap_ci_auc()            - Bootstrap confidence intervals
  calculate_cohens_d()          - Effect size calculation
```

- ASSESSMENT: Well-designed, reusable statistical framework

6.2 utils_model_loader.R (~280 lines)

```
KEY FUNCTIONS:
  find_artifact()     - Detect naming conventions (CL07_* or current)
  load_tokenizer_auto() - Load Keras tokenizer
  load_all_artifacts() - Load all required artifacts
  find_all_models()    - Scan for available models
```

- ASSESSMENT: Enables backward compatibility with Jihad's artifacts

7. Overall Code Quality Assessment

7.1 Strengths

- Well-documented with clear comments and section headers
- Modular design with reusable utility functions
- Comprehensive error handling and validation
- Follows consistent coding style throughout
- Produces publication-ready outputs (tables, figures)
- Rigorous statistical methodology (permutation tests, FDR)
- Backward compatible with original Knox/Obeid artifacts

7.2 Technical Considerations

- Memory: Loading 10 CNN models requires significant RAM
- Compute: 10,000 permutations per metric is time-intensive
- Dependencies: Requires TensorFlow, Keras, quanteda packages
- Data: Assumes specific column naming conventions

7.3 Reproducibility

- Environment: Conda setup script provided (setup_environment.sh)
- Seeds: Random seeds set for reproducible results
- Outputs: All results saved to CSV/RDS for verification
- Documentation: Comprehensive README and method guides

8. Summary

8.1 Paper Readiness

The repository contains all necessary components for a high-quality research publication: rigorous methodology, comprehensive results, statistical validation, and interpretable findings. The existing AMIA_Paper_Methodology_Results_Discussion.md provides ~12,000 words of publication-ready content.

8.2 Pipeline Quality

The code demonstrates professional software engineering practices with clear documentation, modular design, and comprehensive testing. The pipeline is production-ready for research use, with appropriate caveats about computational requirements.

8.3 Key Findings

- CNN achieves excellent performance (AUC=0.987)
- No significant demographic bias detected
- Features are clinically interpretable and consistent
- Three-dimensional bias framework provides comprehensive analysis

8.4 Recommended Next Steps

- External validation at additional sites
- Prospective clinical deployment study
- Expansion of Hispanic/Asian subgroup samples
- Integration into clinical decision support workflow